1)Create a table listing 5 common network threats (such as phishing, DDoS, man-in-the-middle, malware, and password attacks), and for each, write one real-world example from an app or service you use (like WhatsApp, Instagram, or Paytm).

Ans. 5 Common Network Threats with Real-World Examples

No.	Network Threat	Real-World Example
1.	Phishing	Someone sends a fake Instagram login link to steal a user's password.
2.	DDoS Attack	A website like Paytm becomes unavailable because it receives too much unwanted traffic.
3.	Man-in-the-Middle Attack	A hacker tries to read data when someone uses an unsecured public Wi-Fi network to access WhatsApp Web.
4.	Malware	A user downloads a harmful file that can affect their phone while using Instagram.
5.	Password Attack	Someone tries to guess a user's Flipkart account password to access their account.

2)Simulate a brute-force attack scenario by writing a simple Python script that tries to guess a 4-digit PIN code by checking all possible combinations against a preset value. The script should print the number of attempts it took to guess the correct PIN.

Ans. **Python Program: Brute-Force Attack Simulation**

This program tries different 4-digit PIN numbers until it finds the correct PIN.

Code

```
correct_pin = "1234"
attempts = 0

for i in range(10000):
    guess = str(i).zfill(4)
    attempts += 1
```

```
    if guess == correct_pin:
        print("Correct PIN:", guess)
        print("Total Attempts:", attempts)
        break
```

Output

```
Correct PIN: 1234
Total Attempts: 1235
```

Explanation

1. First, we set the correct PIN as `1234`.
2. The program tries different PIN numbers.
3. It counts every attempt.
4. When the correct PIN is found, the program stops.
5. Finally, it displays the PIN and total attempts.

3)List 4 physical security risks that could affect a server room hosting a popular app like Zomato or BookMyShow, and suggest one mitigation step for each risk.

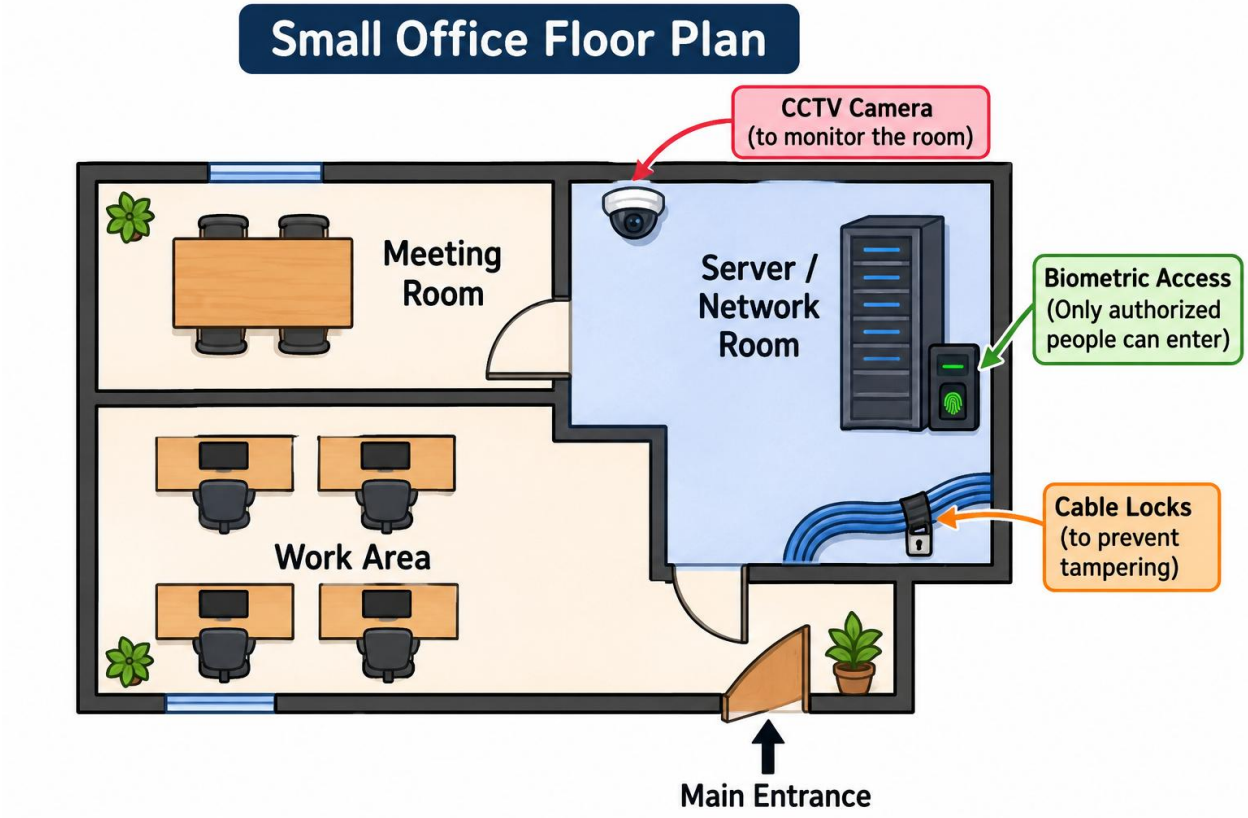
Ans. 4 Physical Security Risks in a Server Room

A server room stores important computers and data. Apps like Zomato and BookMyShow need server rooms to keep their services running.

No.	Physical Security Risk	Mitigation (Solution)
1.	**Unauthorized Access: Unknown people may enter the server room.**	**Use CCTV cameras, security guards, and biometric locks.**
2.	**Fire: Fire can damage servers and other equipment.**	**Install fire alarms and fire suppression systems.**
3.	**Power Failure: Electricity problems can stop servers from working.**	**Use UPS and backup generators.**
4.	**Water Leakage: Water can damage computers and servers.**	**Install water leak detectors and keep servers away from water pipes.**

4)Draw a simple floor plan (hand-drawn or digital) of a small office and mark at least 3 physical security measures you would implement to protect network equipment (such as CCTV, biometric access, or cable locks).

Ans.



5)Use ChatGPT or a similar AI tool to generate a checklist of 7 actions a network admin should take to mitigate risks from both network threats and physical security breaches, then review and edit the checklist to make it specific for a college computer lab environment.

Ans.

College Lab Security Checklist

1. **Secure physical hardware with cable locks and room access**

   - **Bolt computers to desks and keep the lab locked when staff isn't present to prevent theft or tampering.**

2. **Use reboot-to-restore software (like Deep Freeze)**

   - **Ensures any malware, downloaded files, or system changes made by students are wiped clean every time the PC restarts.**

3. **Isolate the lab network using VLANs**

   - **Place lab computers on a separate network segment so students cannot access sensitive campus administration or financial servers.**

4. **Require unique individual student logins**

- **Avoid shared generic guest accounts so you can track exactly who was logged into a machine if a security incident occurs.**

5. **Restrict USB ports and external storage devices**

- **Use Group Policy to block unauthorized flash drives or external hard drives that could carry malware or steal data.**

6. **Disable or secure unused Ethernet wall ports**

- **Prevent unauthorized devices or rogue routers from being plugged directly into the active campus network.**

7. **Automate updates and monitor network traffic**

- **Set up automatic security patches for operating systems and software, and keep an eye out for unusual bandwidth spikes.**